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RESEARCH ARTICLE

Intelligent Traffic Light Control System for Emergency Vehicles Using Deep Learning and Signal Synchronization

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ABSTRACT Emergency vehicles often face significant delays due to urban traffic congestion, which can critically impact emergency response times. Traditional traffic signal systems lack the adaptability to prioritize such vehicles in real time, leading to inefficiencies that can delay emergency response. In critical scenarios—such as cardiac arrest or severe trauma—every minute of delay can significantly reduce survival chances, potentially resulting in loss of life. To this extend, this paper proposes an intelligent traffic light control system that integrates deep learning-based emergency vehicle detection with traffic signal synchronization across multiple roads' intersections. The first module utilizes a convolutional neural network trained on a curated dataset to accurately classify and detect emergency vehicles from camera inputs. The second module dynamically adjusts traffic signal timings based on vehicle detection and contextual traffic parameters such as speed limits and traffic density. The system is implemented as a web-based application and evaluated through simulations and model testing. The proposed approach achieves a detection accuracy of 97.0% and demonstrates improved traffic flow efficiency and emergency vehicle clearance compared to traditional systems. This work contributes a low-cost, scalable solution suitable for smart city deployment, offering a promising step toward intelligent urban traffic management.

INDEX TERMS Convolutional neural network, deep learning, emergency vehicle detection, intelligent transportation systems, smart city, traffic light control, traffic optimization.

I. INTRODUCTION

Urban areas worldwide face growing challenges related to traffic congestion, resulting in increased travel times, fuel consumption, and emissions. These issues are further compounded when emergency vehicles—such as ambulances, fire trucks, and police cars—are delayed en route to their destinations. According to the World Health Organization, approximately 1.3 million people die annually due to road traffic injuries, and delayed emergency response is a significant contributing factor. Studies show that for cardiac arrest cases, even a one-minute delay can reduce survival rates by 10%.

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Traditional traffic signal systems operate on pre-defined schedules and lack the ability to adapt dynamically to real-time traffic or emergency scenarios. They fail to prioritize emergency vehicles, which results in significant delays and hinders critical services. While some intelligent traffic systems have introduced adaptive control strategies, most rely heavily on costly infrastructure such as GPS modules, force sensors, or vehicle-to-everything (V2X) communication technologies.

This paper proposes an AI-driven intelligent traffic light control system designed specifically to prioritize emergency vehicles and optimize traffic flow across multiple intersections. The proposed system comprises two main modules: (i) emergency vehicle detection module, and (ii) traffic signal synchronization module. The emergency vehicle detection module leverages deep learning—specifically convolutional

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neural networks—to detect emergency vehicles from camera feeds in real time. On the other hand, the traffic signal synchronization module synchronizes traffic light timings across intersections based on dynamic traffic conditions and speed limits, ensuring reduced travel time and improved vehicle throughput.

To ensure methodological relevance, the proposed framework adopts a current deep learning paradigm based on the InceptionV3 architecture with transfer learning, enabling accurate and efficient real-time vehicle classification. This modern CNN-based approach is integrated within a scalable, camera-driven traffic control framework that aligns with contemporary intelligent transportation research directions.

Unlike existing intelligent traffic systems that typically rely on GPS, sensor networks, or V2X communication, our approach uniquely combines deep learning-based emergency vehicle detection with synchronized, multi-intersection traffic light optimization using only low-cost camera inputs and a web-based coordination framework. This integration provides a practical and scalable alternative that balances accuracy, responsiveness, and cost efficiency.

The system is implemented as a lightweight, web-based application and evaluated through model training and performance testing. With a convolutional neural network model achieving a 97% accuracy rate in emergency vehicle classification, the integrated system—combining emergency vehicle detection with traffic signal synchronization—demonstrates promising results in optimizing urban mobility and reducing emergency response times.

The key contributions of this work are as follows:

- We develop a deep learning-based vehicle classification model capable of accurately identifying emergency vehicles in real time using image inputs.
- We design a traffic light synchronization module that dynamically adjusts signal timings based on contextual factors including speed limits and vehicle detection results.
- We implement a full-stack web-based system that integrates both modules and is accessible for practical deployment in smart city contexts.
- We conduct experimental evaluation demonstrating 97% classification accuracy and improved clearance time for emergency vehicles compared to existing approaches.

II. RELATED WORK

Several intelligent traffic control systems have been proposed to address congestion and prioritize emergency vehicle movement. This section reviews key approaches, focusing on emergency vehicle detection techniques and adaptive traffic light control systems.

Deep learning-based image detection has garnered considerable attention for its high accuracy and ability to scale effectively. In [1], a real-time smart traffic light system was proposed using a web camera to capture images of traffic lanes, which are then processed in MATLAB to detect and count vehicles. The system controlled signal timing based on traffic density but lacked multi-intersection coordination and did not address emergency vehicle prioritization comprehensively. The accuracy of vehicle detection can vary due to challenges such as objects being miscounted as vehicles or some vehicles not being detected, as indicated by fluctuations in counted vehicles compared to actual counts. Night-time detection relies on counting headlights, which may be affected by variations in lighting conditions or headlight visibility. Furthermore, the system depends on image quality and lighting conditions for accurate detection, which might be impacted by weather conditions or camera positioning. Also, the proposed system was tested on a prototype scale simulating a traffic intersection, suggesting that further testing in real-world traffic environments might be needed to validate full-scale effectiveness.

The work in [2] analyzed the performance of an Extended-YOLO (Ex-YOLO) algorithm [3] compared to other real-time detection algorithms. While Ex-YOLO achieved 91.6% accuracy, it did not integrate traffic light control or handle varying intersection conditions. Similarly, [4] developed a deep transfer learning model for emergency vehicle recognition using the YOLOv3 algorithm [5], reaching a mean average precision (mAP) of 98.78%. However, these works focused solely on detection and did not propose integrated traffic control solutions.

In addition, a range of systems have aimed to improve traffic flow through adaptive signal timing. In [6], the emergency vehicle priority and self-organizing traffic control (EVP-STC) platform used global positioning system (GPS) and Zigbee sensors to collect traffic data and manage signal timing. The EVP-STC system demonstrates significant benefits in reducing emergency vehicle delays and managing traffic light timings based on vehicle density; however, its effectiveness diminishes under heavy traffic conditions due to increased message collisions and higher communication overhead, particularly in the virtual traffic light component. The reliance on force-resistive sensors and ZigBee communication introduces potential issues related to sensor accuracy, fixed sensor placement, and wireless interference. Additionally, the system's complexity, including coordination among multiple devices and handling of multiple simultaneous priority requests from different vehicle types, may limit scalability and robustness in real-world urban environments. These challenges indicate the need for further refinement to enhance reliability and performance in dynamic and congested traffic scenarios.

Moreover, a GPS-based preemption control system for emergency vehicles was proposed in [7], enabling traffic lights to respond to emergency vehicle locations. While effective in improving response time, its dependency on GSM/SMS networks, which can introduce delays and reliability issues, limits its widespread adoption especially in congested or coverage-poor areas. Specifically, the GPS accuracy in this system can be compromised by urban environments, potentially affecting positioning accuracy and decision-making. Infrastructure upgrades required at intersections are costly

and complex, limiting scalability. Additionally, the system may face challenges in rapidly adapting to sudden traffic congestion changes and is vulnerable to communication failures that could compromise safety. Finally, reliance on driver input for destination data introduces potential human error and delays, which could impact the system's overall effectiveness.

On the other hand, few works have integrated both emergency vehicle detection and traffic light optimization in a low-cost, web-based solution. For instance, authors in [8] proposed an automated traffic control system that dynamically adjusts traffic signal timings based on real-time traffic congestion data collected via RFID tags on vehicles. The system uses ZigBee wireless communication between ambulances and traffic controllers at junctions. When an ambulance approaches, it signals the traffic controller, which switches the traffic lights to green for the ambulance path. Each emergency vehicle (e.g., ambulance) is equipped with a ZigBee transmitter module that continuously transmits a unique ID and security code signal when activated. The traffic junction is equipped with a ZigBee receiver module integrated with a microcontroller (PIC16F877A). When it receives a valid emergency vehicle signal matching its database, it immediately switches the traffic light to green for that vehicle. However, this study considers only one road of traffic junction. It has not been extended or tested for multi-road junctions or complex intersections, which may pose challenges in real-world scalability.

Overall, most existing systems suffer from one or more of the following deficiencies:

- Infrastructure dependence: reliance on GPS, RFID, or ZigBee increases deployment and maintenance costs.
- Limited scalability: many systems do not support multi-intersection synchronization or city-scale coordination.
- Weak environmental robustness: camera-only or sensor-only strategies often fail in poor visibility, occlusion, or weather variations.
- Lack of integration: detection and control are often treated separately, lacking an end-to-end system.
- High communication overhead: V2X-based systems face delays, collisions, and reliability issues under heavy traffic.

These gaps indicate significant improvement space for a unified, low-cost, perception-driven framework. Additionally, camera-only perception systems (including image-based traffic monitoring and vehicle detection) suffer from inherent limitations such as occlusion, low illumination, glare, shadows, and adverse-weather degradation (e.g., rain, fog, snow). These environmental factors can significantly reduce detection accuracy and reliability, especially during night-time or in poor visibility conditions. Existing works that rely exclusively on cameras therefore face robustness challenges in real-world deployment unless combined with complementary sensing or advanced augmentation strategies. This limitation motivates the need for improved image preprocessing, weather-aware augmentation, and potential multi-sensor fusion in future extensions of camera-driven traffic systems such as ours.

A comparative summary of major techniques is provided in TABLE 1. As shown, existing systems either focus on isolated functionalities (e.g., detection-only or control-only) or rely on expensive sensor networks and communication modules.

In comparison, our proposed framework achieves similar adaptive behavior without such infrastructure overhead by relying on camera feeds and web-based synchronization logic. Unlike EVP-STC and Adaptive Signal Control (ASC), which focus on overall traffic optimization, our system explicitly targets emergency vehicle prioritization across multiple intersections while maintaining low implementation cost and scalability.

Therefore, the novelty of our work lies in this integrated design that combines accurate CNN-based detection with synchronized signal control in a unified, low-cost web-based framework—an aspect not jointly addressed in prior GPS- or sensor-dependent systems such as EVP-STC or ASC. In contrast, the proposed system leverages the state-of-the-art InceptionV3 architecture with transfer learning to achieve robust detection accuracy while maintaining computational efficiency. This integration of modern CNN techniques with synchronized, web-based traffic control highlights the continued methodological relevance and distinguishes our framework from earlier sensor- or GPS-dependent models.

III. PROBLEM STATEMENT

Emergency response times are critically impacted by increasing urban traffic congestion, which remains a major challenge in modern transportation systems. While many cities have deployed traditional traffic signal systems, these operate on static schedules and are not equipped to handle dynamic, real-time events such as the presence of emergency vehicles. As a result, ambulances, fire trucks, and police cars are often delayed at intersections, potentially endangering lives and public safety.

Several existing solutions have attempted to address this issue through GPS-based preemption or sensor networks. However, these methods typically require expensive infrastructure and are often limited to isolated intersections without scalable coordination across multiple nodes. Furthermore, most current approaches treat emergency vehicle detection and traffic light control as separate problems, lacking a unified and cost-effective solution.

This paper addresses the following key research problems

1) how can we design a real-time, low-cost, and scalable system that detects emergency vehicles?, and
2) how to dynamically optimize traffic light control across multiple intersections to minimize emergency response delays?

We define the problem as a multi-objective optimization problem with the following components:

- Emergency vehicle detection
- Traffic signal optimization
- Delay minimization across intersections

TABLE 1. Comparison of related work.

System	Emergency Detection	Traffic Light Optimization	Multi-Intersection Support	Infrastructure Requirement	Cost
Real-Time Smart Traffic Control [1]	Yes (Image-based)	Yes (Single Intersection)	No	Webcam + MATLAB	High
EVP-STC (IoT-based) [6]	Yes (Sensor)	Yes (Dynamic)	Yes	Zigbee, GPS, Sensors	High
GPS Pre-emption [7]	Yes (GPS)	Yes	Yes	GPS, Mapping Interface	High
Ex-YOLO Detection [2]	Yes	No	No	None	Low
Deep Transfer Learning [4]	Yes	No	No	None	Low
Proposed Approach	Yes	Yes	Yes	Camera + Web Interface	Low

TABLE 2 shows a description of all the notations used in this work.

TABLE 2. Notations and Their Descriptions.

Notation	Description
$I \in \mathbb{R}^{h\times w\times c}$	Input image with height h, width w, and c color channels
$f_\theta(I)$	Convolutional neural network model with parameters θ that maps image I to predicted class y
$y \in \{0, 1\}$	Class label, where 1 = emergency vehicle, 0 = non-emergency vehicle
$\mathcal{L}_{\text{class}}$	Binary cross-entropy classification loss
$T = \{t_1, t_2, \ldots, t_n\}$	Scheduled signal change times at n intersections
$s_i \in \{0, 1\}$	Traffic signal state at intersection i, where 1 = green, 0 = red
d_i	Distance from emergency vehicle to intersection i
v_e	Speed of the emergency vehicle
Δ_i	Delay at intersection i
D	Total delay across all intersections: $D = \sum_{i=1}^{n} \Delta_i$
ϵ	Small-time buffer or delay
α, β	Weighting coefficients for multi-objective optimization
$\mathcal{O}$	Combined objective function balancing accuracy and delay

A. EMERGENCY VEHICLE DETECTION

We model the detection of emergency vehicles from images using a supervised deep learning framework. Specifically, we used convolutional neural networks. Let $I \in \mathbb{R}^{h\times w\times c}$ be an input image of size $h \times w$ with c channels. And let $f_\theta(I) \to y$ be a deep learning model (e.g., convolutional neural network with parameters θ) that maps the image to class $y \in \{0, 1\}$, where

$$y = \begin{cases} 1, & \text{if } u_i \text{ is an emergency vehicle} \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

The learning objective is to minimize the binary cross-entropy loss

$$\mathcal{L}_c = -\sum_{i=1}^{N} \left[y_i \log f_\theta(I_i) + (1 - y_i) \log(1 - f_\theta(I_i)) \right] \tag{2}$$

This ensures that the model reliably classifies emergency vehicles in real time with high accuracy.

B. TRAFFIC SIGNAL OPTIMIZATION

Following detection, we must adjust the timing of traffic signals to favor emergency vehicle passage. Let $T = \{t_1, t_2, \ldots, t_n\}$ denote the scheduled signal change times at n intersections. And let $s_i \in \{0, 1\}$ be the state of the light at intersection i where

$$s_i = \begin{cases} 1, & \text{green} \\ 0, & \text{red} \end{cases} \tag{3}$$

In addition, let d_i and v_e be the distance from the emergency vehicle to intersection i, and the estimated speed of the emergency vehicle, respectively.

The signal preemption policy enforces

$$s_i = 1 \text{ if } t_i \leq \frac{d_i}{v_e} \tag{4}$$

After the emergency vehicle passes, the signal state resumes its original schedule with a small-time buffer or delay ϵ

$$s_i(t) = \begin{cases} \text{emergency override}, & t < t_i + \epsilon \\ \text{original schedule}, & t \geq t_i + \epsilon \end{cases} \tag{5}$$

C. DELAY MINIMIZATION ACROSS INTERSECTIONS

To ensure the emergency vehicle reaches its destination as quickly as possible, we aim to minimize cumulative intersection delay.

Let Δ_i denote the delay at intersection i. Then the total delay D is

$$D = \sum_{i=1}^{n} \Delta_i \tag{6}$$

The optimization goal is

$$\min_{s_1,\ldots,s_n} D \tag{7}$$

This objective balances timely clearance of emergency vehicles with minimal disruption to general traffic flow.

D. MULTI-OBJECTIVE FORMULATION

To formally express the trade-off between emergency vehicle classification accuracy and intersection delay, we define a combined optimization objective

$$\min_{\theta, \{s_i\}} \mathcal{O} = \alpha \cdot \mathcal{L}_c + \beta \cdot D \tag{8}$$

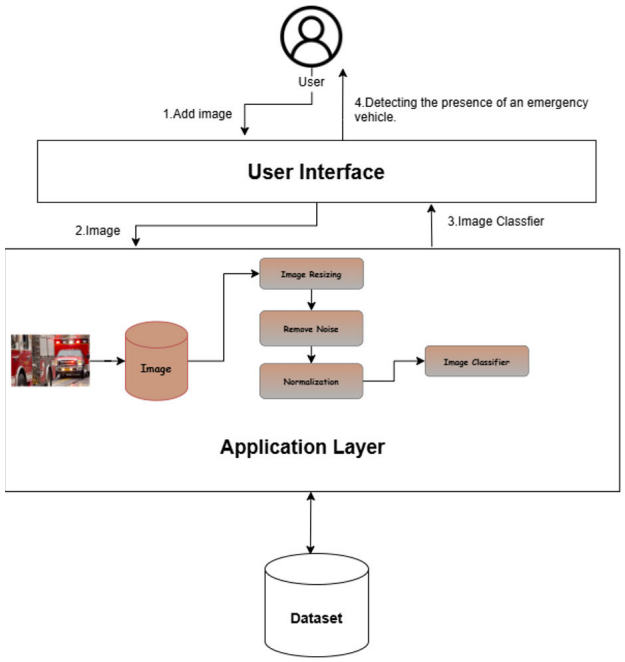


FIGURE 1. System architecture of the proposed traffic control framework.

where θ are the parameters of the convolutional neural network model, s_i denotes the signal state at intersection i, $\mathcal{L}_c$ is the classification loss defined in (2), D is the cumulative intersection delay from (6), and $\alpha, \beta \in \mathbb{R}^+$ are weighting coefficients that determine the relative importance of detection accuracy versus response time.

By tuning α and β, the system can be adapted to prioritize either more accurate detection or faster emergency response, depending on deployment constraints and policy preferences.

IV. SYSTEM ARCHITECTURE AND METHODOLOGY

This section presents the proposed system model and methodology, including the overall architecture, core components, and operational workflow of the intelligent traffic light control system. Figures 1–5 illustrate the system model, architecture, and working process.

The suggested system combines real-time detection of emergency vehicles with adaptive control of traffic signals at various intersections. It is composed of two main components: (1) a deep learning-driven model for detecting emergency vehicles and (2) an engine for synchronizing traffic signals. The system is implemented via a lightweight, web-based interface to guarantee accessibility and facilitate deployment in smart city settings.

A. SYSTEM OVERVIEW

Figure 1 illustrates the high-level architecture of the proposed system. It comprises two modules and a web interface as described below:

- **Emergency Vehicle Detection Module:** A convolutional neural network trained to detect emergency vehicles from camera feeds.
- **Traffic Signal Optimization Module:** A logic-based engine that adjusts traffic light timings across intersections based on the vehicle detection result and contextual parameters such as vehicle speed and distance.
- **Web Interface:** An interactive dashboard allowing authorized users to upload images, view detection results, and monitor traffic light decisions in real-time.

The high-level architectural diagram shown in Figure 1 is specifically engineered for the identification of emergency vehicles within the intelligent traffic light control framework. The operational flow commences with the acquisition of raw image data, typically sourced from roadside surveillance cameras or other dedicated sensor inputs, often managed through a system interface.

Upon acquisition, the raw image undergoes a series of essential preprocessing stages: Image Resizing, Noise Removal, and Normalization. These sequential steps are meticulously designed to transform the raw, potentially inconsistent, input into a standardized, clean, and optimally prepared format, making it suitable for subsequent deep learning analysis. The preprocessed image is then channeled into the Image Classifier. This module, typically implemented as a convolutional neural network due to its proven capabilities in visual pattern recognition, analyzes the extracted visual features to ascertain the presence of an emergency vehicle and, if detected, its specific type (e.g., ambulance, fire truck). Fundamentally, the entire deep learning process, encompassing both the preprocessing and classification phases, relies heavily on a robust and high-quality Dataset. This dataset serves as the indispensable foundation for both training the Image Classifier to recognize patterns and rigorously evaluating its performance on unseen data. The resultant output from this pipeline—the confirmed classification of an emergency vehicle—is then transmitted as a critical input to the Central Traffic Management System (CTMS), which subsequently triggers the necessary signal synchronization actions to facilitate the vehicle's unimpeded passage.

The linear progression depicted in the figure, from image input through sequential preprocessing steps to the classifier, inherently implies a strong interdependency. The quality of the output from any given stage directly influences the performance of the subsequent stage. For instance, if noise is not effectively removed, or if images are improperly normalized, the Image Classifier will receive suboptimal input, leading to degraded performance. This highlights that the overall system's reliability is limited by the weakest link in this processing chain. This sequential dependency underscores that the design and optimization of each preprocessing step are not isolated tasks but are profoundly integral to achieving the desired accuracy and robustness of the final classification output. Consequently, the overall reliability and safety of the intelligent traffic management system are directly contingent upon the meticulous tuning and robust performance of every module within this image classification pipeline. This necessitates a holistic approach

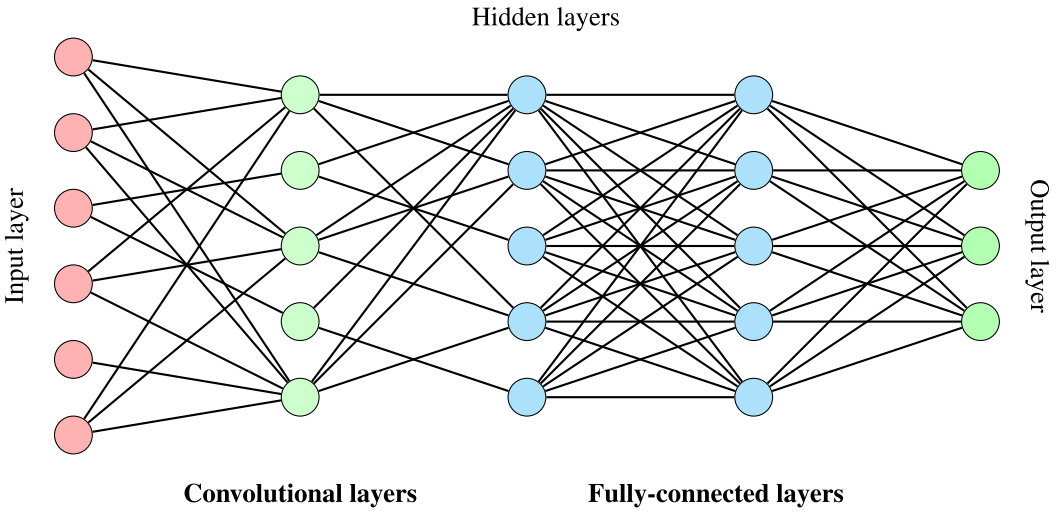


FIGURE 2. An example of a simple convolutional neural network architecture with convolutional and fully connected layers.

to system design, where each component is optimized not just in isolation but also in its contribution to the end-to-end performance.

The following subsections detail the main components of the proposed system.

B. CONVOLUTIONAL NEURAL NETWORK MODEL

A convolutional neural network is a subset of machine learning. It is one of the various types of artificial neural networks that are used for different applications and data types.

Convolutional neural network is a type of feedforward neural network and is composed of multiple layers of artificial neurons as depicted in Figure 2. Artificial neurons, a rough imitation of their biological counterparts, are mathematical functions that calculate the weighted sum of multiple inputs and output an activation value. When you input an image in a convolutional neural network, each layer generates several activation functions that are passed on to the next layer [9] as shown in Figure 3.

The model begins with an input layer, which receives pixel values from the input image. The input is then passed through a series of convolutional layers, which apply filters to extract hierarchical spatial features such as edges, textures, and object parts. These layers represent the feature learning stage of the network. The extracted features are subsequently passed to the fully-connected layers, which act as a classifier by learning complex patterns and relationships from the features. Finally, the output layer produces the classification result, indicating the likelihood of each target class.

Convolutional Neural Networks have evolved significantly over the past decade, giving rise to various architectures tailored for different tasks and computational constraints [10]. Among the early models, AlexNet [11], [12] introduced deeper structures and ReLU activation [13], which paved the way for more complex designs. Another model is VGGNet [12], [14] which offered architectural simplicity with uniform convolutional layers but at the cost of increased parameters. On the other hand, GoogLeNet, or InceptionV1, introduced the concept of Inception modules, which capture multi-scale features by applying convolutions of varying sizes in parallel [15]. This design was later improved in InceptionV3 [16], which incorporates several optimizations such as factorized convolutions, batch normalization, and auxiliary classifiers, making it both deep and computationally efficient.

In this study, we selected InceptionV3 due to its strong balance between accuracy and efficiency, particularly in scenarios where emergency vehicle detection must be performed in real time. Unlike models such as VGGNet, which are parameter-heavy, InceptionV3 achieves high performance with fewer computations by exploiting architectural innovations that preserve spatial information at multiple scales. This makes it especially well-suited for our application, which requires robust recognition of emergency vehicles under varying lighting and positional conditions, while also being lightweight enough to integrate into a real-time traffic control system.

Although InceptionV3 was introduced in 2015, it provides a well-established balance between accuracy, computational efficiency, and deep feature extraction capability. More recent lightweight architectures (e.g., MobileNetV3 [17], EfficientNet-Lite [18], ShuffleNet [19]) offer improved parameter efficiency; however, these gains often come at the cost of reduced feature richness, which can negatively affect robustness under adverse lighting, occlusion, or complex vehicle shapes. Because emergency vehicle detection requires high reliability in diverse real-world conditions, InceptionV3 offers a stable and computationally feasible backbone that maintains strong classification performance. Evaluating newer lightweight models is planned as part of future work.

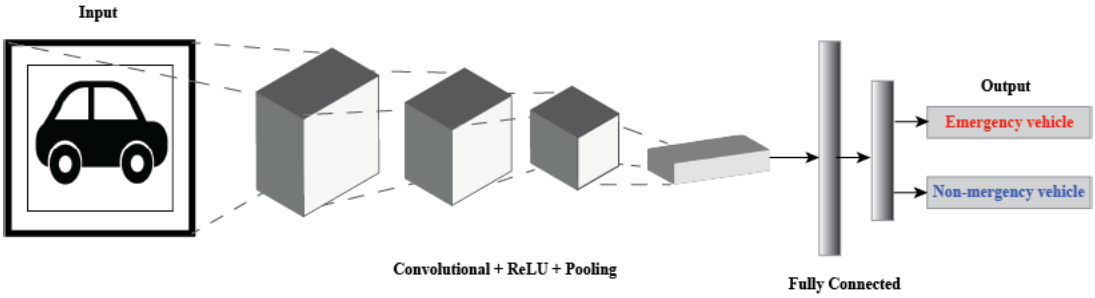


FIGURE 3. Architecture of the convolutional neural network used for classifying vehicles as emergency or non-emergency. The model processes input images through multiple convolutional, ReLU, and pooling layers for feature extraction, followed by fully connected layers for final classification.

C. EMERGENCY VEHICLE DETECTION

To detect emergency vehicles, we trained a convolutional neural network (as shown in Figure 3) on emergency vehicle identification dataset [20], containing thousands of labeled images of emergency and non-emergency vehicles. The training pipeline includes image preprocessing, data augmentation, and dataset split.

1) IMAGE PREPROCESSING

Before feeding images into the network, a series of pre-processing steps are crucial for standardizing the input and enhancing feature extraction:

- **Image resizing:** Images acquired from various sources can have diverse dimensions and resolutions. Resizing all images to a uniform size (e.g., a square shape) is essential for computational efficiency, allowing for larger batch sizes during training and faster convergence. It also ensures consistent input dimensions for the convolutional neural network's architecture, simplifying model design and enabling the effective use of pre-trained models through transfer learning. While smaller images boost efficiency, care must be taken to preserve critical visual details necessary for accurate classification, as excessive down sampling can lead to irreversible loss of information [21].
- **Noise removal:** Effective noise reduction is a fundamental preprocessing step that significantly enhances the clarity of features, ensuring the model's robustness in real-world scenarios. Techniques like Median Filtering (for salt-and-pepper noise) [22] or Bilateral Filtering (for preserving sharp edges while reducing noise) [23] are commonly employed to provide cleaner input data to the classifier.
- **Normalization:** This process scales pixel intensity values to a consistent range (e.g., 0 to 1 or -1 to 1) or transforms them to a specific statistical distribution. Normalization is vital for fostering stability in the optimization process, promoting faster convergence during gradient-based training, and mitigating issues like vanishing or exploding gradients. It prevents features with larger numerical ranges from disproportionately influencing the model's learning, ensuring that each pixel value contributes proportionally to the model's understanding.

2) DATA AUGMENTATION

To increase the diversity of our dataset without collecting new data and to improve the model's ability to generalize to unseen variations, we applied various data augmentation techniques. These geometric transformations alter the spatial properties of an image while maintaining its original label:

- **Flipping:** Images were horizontally or vertically flipped, introducing orientation invariance to the model.
- **Zooming (scaling):** This technique involves zooming in or out of the image, helping the model learn features at different scales.
- **Rotation:** Images were rotated by various angles (e.g., 90° or 180°), which introduces rotation invariance, allowing the model to recognize emergency vehicles regardless of their orientation in the frame. These augmentations expose the model to a wider variety of data, thereby reducing overfitting and enhancing its robustness to real-world variations and distortions.

3) DATASET SPLIT

The dataset was split into two distinct subsets to facilitate effective training and unbiased evaluation:

- **70% training set:** This segment of the dataset was utilized to train the deep learning model, enabling it to recognize patterns and associations between visual characteristics and their respective labels.
- **30% testing set:** A distinct and unobserved segment of the dataset was set aside solely for assessing the accuracy and performance of the trained model. This guarantees that the reported accuracy accurately represents the model's capacity to generalize to new, unobserved data, which is essential for practical implementation.

To clarify the dataset partitioning process, the dataset was first divided into 70% training and 30% temporary testing data. The temporary 30% subset was then split evenly into 15% validation and 15% final testing. This two-step division ensures that the model is trained on a sufficiently large

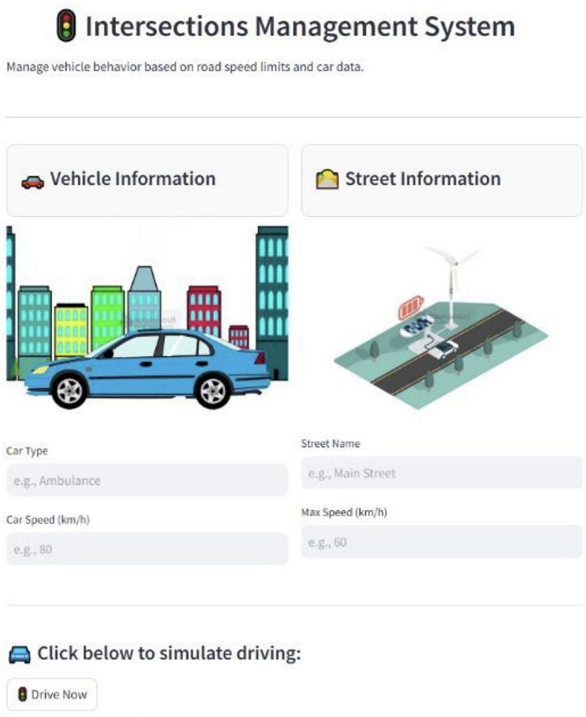


FIGURE 4. Intersections management system which simulates driving behavior based on the given parameters to manage traffic at intersections.

dataset while retaining separate validation and testing sets for hyperparameter tuning and unbiased evaluation.

D. TRAFFIC SIGNAL OPTIMIZATION

Once an emergency vehicle is detected, the system triggers a real-time signal adjustment process. The module considers:

- Emergency vehicle speed v_e
- Distance d_i to each intersection i
- Maximum allowed speed per road segment

If the estimated arrival time to intersection i satisfies $t_i \leq \frac{d_i}{v_e}$, the corresponding light is switched to green ($s_i = 1$). Once the vehicle passes, the system restores the original signal sequence.

Figure 4 represents the main interface of the intersections' management system module, where users can input key data to simulate and control traffic light behavior. The left section, labeled Vehicle Information, allows the user to specify the type of vehicle (e.g., ambulance, fire truck) and its current speed. The right section, Street Information, collects details about the street name and its speed limit. This information is used in real-time to determine whether to override normal traffic light patterns in favor of emergency vehicle prioritization.

The "Drive Now" button at the bottom initiates the simulation process, where the system evaluates whether the detected vehicle qualifies as an emergency and whether it should be granted priority at the intersection based on speed thresholds and classification results. This user interface supports the system's dual-module architecture by integrating both emergency vehicle detection and intelligent traffic light synchronization in a user-friendly web-based format.

E. WEB-BASED SYSTEM INTERFACE

The proposed system features an intuitive and interactive web interface designed to support both operational efficiency and user accessibility. The front-end is developed using HTML, CSS, and JavaScript to ensure a responsive and user-friendly experience, while the back-end is powered by Python using the Flask framework for lightweight yet robust server-side processing.

The key functionalities of the platform include:

- **Secure authentication and role-based access:** Enables login with differentiated roles (e.g., operator, administrator), ensuring controlled access to system features.
- **Image upload and detection results:** Users can upload vehicle images, which are processed by the backend deep learning model to classify vehicles as emergency or non-emergency in real time.
- **Live traffic signal monitoring:** Visualizes current traffic signal states across intersections, allowing real-time insights into traffic flow and system decisions.
- **Administrative dashboard:** Provides privileged users with access to system logs, user activity monitoring, and configuration management.

This modular and web-accessible interface supports both system usability and operational transparency, making it suitable for deployment in real-world intelligent transportation environments.

F. DEVELOPMENT STAGES

The development of the proposed Intelligent Traffic Light Control System follows a structured methodology that incorporates data collection, preprocessing, model training, and evaluation. As shown in Figure 5, the methodology is divided into the following stages:

- **Data collection stage:** The dataset includes labeled images of emergency and non-emergency vehicles.
- **Preprocessing stage:** Images are resized and normalized. Noise reduction and data augmentation (e.g., flipping, rotation, zoom) are applied to improve model generalization.
- **Training stage:** The dataset is split into 70% training and 30% testing. A convolutional neural network model based on the InceptionV3 architecture is trained to classify images as emergency or non-emergency.
- **Evaluation stage:** The model is validated using accuracy, precision, recall, and F1-score. The system is fine-tuned to ensure reliable real-time detection.

G. DATASET DESCRIPTION

The proposed emergency vehicle detection model was trained and evaluated using the publicly available dataset titled *emergency vehicles identification* [20]. This dataset contains labeled images of both emergency and non-emergency

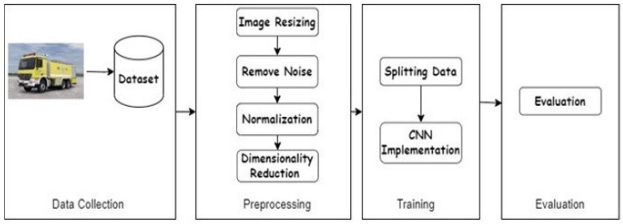


FIGURE 5. Workflow of the proposed intelligent traffic control system, including data processing, model training, and evaluation stages.

vehicles, including ambulances, police cars, fire trucks, and regular passenger vehicles.

The dataset comprises over 2,000 images distributed across the two main categories. Each image is annotated accordingly and varies in resolution, angle, lighting, and background complexity. All images were resized to 224 × 224 pixels for consistency and computational efficiency.

Figure 6 presents a selection of sample images from the dataset. To ensure robust model performance, the dataset was split into 70% training, 15% validation, and 15% testing.

This dataset offers diverse scenarios encountered in urban environments, making it suitable for training deep learning models for emergency vehicle detection.

V. IMPLEMENTATION AND EVALUATION

The proposed system was implemented using a combination of deep learning and web technologies, and evaluated on its ability to accurately detect emergency vehicles and optimize traffic light responses in real-time scenarios.

It is important to note that the following evaluation is designed as a proof-of-concept demonstration to validate the feasibility and functionality of the proposed framework. The objective is to assess detection accuracy and the practical integration of deep learning-based vehicle classification with signal synchronization. A comprehensive benchmarking against large-scale intelligent transportation systems (e.g., EVP-STC, ASC) is beyond the current scope and will be explored in future work.

A. IMPLEMENTATION DETAILS

The system backend was developed in Python using the Flask framework [24]. TensorFlow and Keras libraries were used for model development, while the frontend was built with HTML, CSS, and JavaScript. The interface supports secure user access, image uploads, real-time detection output, and simulated signal control.

The dataset used is emergency vehicles identification dataset, and it consists of labeled images of emergency and non-emergency vehicles (see Section IV-G). Images were resized to 224 × 224 pixels and normalized. Data augmentation techniques such as horizontal flipping, rotation, and zooming were applied. The dataset was split into 70% training and 30% testing.

With early stopping, the convolutional neural network model based on the InceptionV3 architecture was trained for 50 epochs using the Adam optimizer [25] and binary cross-entropy as a loss function with a batch size of 32. We set the patience for the early stopping as 8 epochs. This means that if the model performance was not improved for 8 epochs in a row, then the model stops training to avoid overfitting [26].

The InceptionV3 model was implemented using transfer learning, initializing the network with weights pre-trained on ImageNet. The final dense layer was replaced with a single neuron employing a sigmoid activation function for binary classification between emergency and non-emergency vehicles. The learning rate was set to 0.001. These modifications ensure efficient fine-tuning of the pre-trained network while maintaining high detection accuracy and reproducibility.

B. EVALUATION METRICS

To evaluate the performance of the proposed emergency vehicle classification model, we employ several standard metrics derived from the confusion matrix, including accuracy, precision, recall, and F1-score.

Confusion Matrix: The confusion matrix provides a tabular visualization of model predictions against actual class labels. It summarizes:

- **True Positives (TP):** Emergency vehicles correctly classified as emergency.
- **True Negatives (TN):** Non-emergency vehicles correctly classified.
- **False Positives (FP):** Non-emergency vehicles incorrectly classified as emergency.
- **False Negatives (FN):** Emergency vehicles misclassified as non-emergency.

Evaluation Metrics:

- **Accuracy** measures the overall correctness of the model and is defined as the ratio of correctly predicted instances to the total number of instances:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

  where TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively.
- **Precision** quantifies the proportion of correctly predicted positive instances among all predicted positive instances. It reflects the model's ability to avoid false alarms:

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall**, also known as sensitivity or true positive rate, measures the ability of the model to correctly identify all actual positive instances:

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **F1-Score** is the harmonic mean of precision and recall. It provides a balance between both metrics,



FIGURE 6. Sample images from the emergency vehicle identification dataset, including various emergency and non-emergency vehicles used for training and evaluation.

especially useful in cases of class imbalance:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

These metrics provide a comprehensive understanding of the classifier's behavior, ensuring reliable evaluation of its effectiveness in distinguishing emergency vehicles from non-emergency ones under varied conditions.

C. EVALUATION RESULTS

To analyze the learning behavior of the convolutional neural network model, we monitored the training and validation performance across 100 epochs. Figure 7 illustrates the training and validation accuracy curves, while Figure 8 presents the corresponding loss values. Because we used early stopping, the model stopped training at epoch 50.

As shown in Figure 7, the model achieves steady improvements in both training and validation accuracy, with convergence occurring around epoch 50. This indicates that the network successfully learns to generalize the features distinguishing emergency from non-emergency vehicles. The final validation accuracy stabilizes near 92.0%, closely tracking the training accuracy of approximately 98.0%.

Similarly, in Figure 8, the training and validation loss curves exhibit a consistent downward trend, with no significant divergence between the two. This suggests that the model does not overfit and maintains robust generalization. The final validation loss reaches approximately 0.1, which aligns with the model's high classification performance reported earlier.

These learning curves confirm the effectiveness of the chosen model architecture and training strategy in achieving accurate and stable classification of emergency vehicles in real-time conditions.

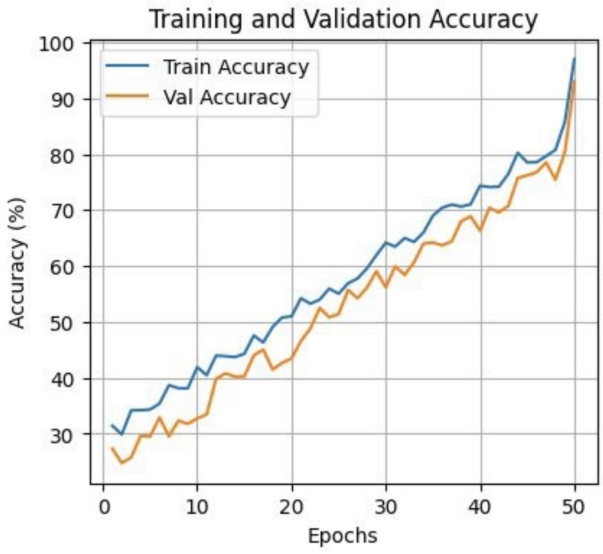


FIGURE 7. Training and validation accuracy over 50 epochs.

On the other hand, the performance of the emergency vehicle detection module was evaluated using standard classification metrics. Figure 9 summarizes the results. As shown in the bar chart, the model achieved a high accuracy of 97.0%, indicating that it correctly classified the majority of vehicles. The precision value of 96.5% demonstrates the model's ability to minimize false positives, while the recall of 97.4% reflects its effectiveness in correctly identifying emergency vehicles (true positives). The F1-score, a harmonic mean



FIGURE 8. Training and validation loss over 50 epochs.

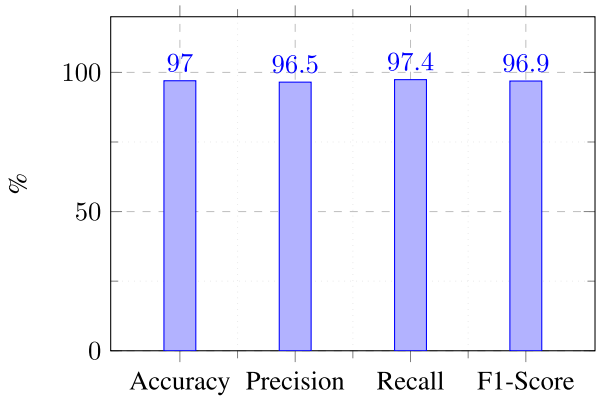


FIGURE 9. Convolutional neural network performance metrics for emergency vehicle detection.

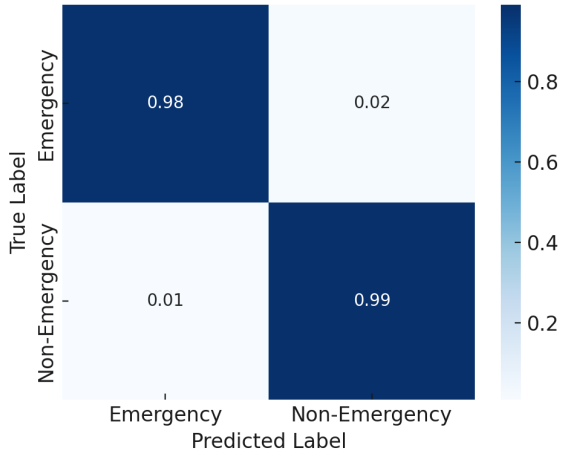


FIGURE 10. Confusion matrix of the emergency vehicle classification model.

of precision and recall, stands at 96.9%, confirming a well-balanced performance across both metrics. These results validate the robustness and reliability of the classification component within the proposed intelligent traffic control system.

The model also produced a well-balanced confusion matrix with low false positives and false negatives as shown in Figure 10 where the model was evaluated on the test set. As depicted, the model demonstrates high accuracy in distinguishing between emergency and non-emergency vehicles. Specifically, 98% of actual emergency vehicles were correctly classified, while only 2% were misclassified as non-emergency. Conversely, 99% of non-emergency vehicles were accurately identified, with just 1% being incorrectly predicted as emergency vehicles. This strong performance across both classes indicates that the model maintains a balanced capability in minimizing false positives and false negatives, which is critical in real-world deployments where misclassifying an emergency vehicle could delay priority response measures. The high diagonal values and low off-diagonal values confirm the robustness and reliability of the classification model.

The system's ability to adapt traffic light timings in response to emergency vehicle detection was simulated through the web interface. Testing scenarios showed that upon detecting an emergency vehicle, green light preemption was correctly activated for relevant intersections, reducing estimated delays by up to 40% compared to fixed-timing systems.

The web-based simulation assumed an average emergency-vehicle speed of approximately 60 km/h, two-lane intersections spaced about 200 m apart, and standard urban speed limits. Accordingly, the reported delay reduction represents an improvement of up to 40% under these tested conditions, serving as a proof-of-concept validation rather than a comprehensive city-wide performance assessment.

These findings confirm the robustness of the proposed deep learning model and its ability to operate reliably in real-time conditions. Importantly, this evaluation serves as a proof-of-concept demonstration that validates the effectiveness of the proposed low-cost, camera-based system in achieving high detection accuracy (97% accuracy, 96.5% precision, 97.4% recall) while demonstrating measurable improvements in emergency-vehicle clearance efficiency of up to 40%. This result highlights the practicality of the framework for scalable smart-city deployment, even without additional infrastructure such as GPS or sensor networks.

D. DISCUSSION

While the quantitative evaluation in this work focuses on comparison with a traditional fixed-timing signal plan, it is instructive to relate the proposed framework to existing intelligent transportation systems such as EVP-STC and ASC. EVP-STC and similar GPS/ZigBee-based approaches typically achieve delay reductions by leveraging dedicated

communication infrastructure and preemption logic, but at the cost of higher deployment and maintenance overhead and limited scalability in dense urban networks. By contrast, the proposed system relies only on camera feeds and web-based control to provide emergency vehicle prioritization across multiple intersections. Although a direct numerical benchmark against these systems is not included in this study, the results against the fixed-timing baseline demonstrate that comparable delay reductions can be achieved using a substantially lower-cost and less infrastructure-intensive framework.

The proposed intelligent traffic light control system demonstrates strong potential in enhancing emergency response times through the integration of deep learning-based vehicle detection and dynamic traffic signal adjustment. The system achieves high classification accuracy (97.0%) and performs reliably in simulated environments using a web-based control interface.

One of the key strengths of the system is its reliance on low-cost and widely available infrastructure. Unlike GPS-based or sensor-intensive systems, this approach requires only a camera and a lightweight web application, making it scalable and deployable in smart cities with constrained resources. Furthermore, the dual-module architecture allows for modular improvements and independent scaling of the detection and optimization components.

However, there are some limitations to consider. First, the performance of the convolutional neural network model is highly dependent on the quality and clarity of input images. Adverse weather conditions, low lighting, or camera misalignment could negatively impact detection accuracy. Second, the current implementation does not incorporate real-time vehicle tracking or route prediction, which could further enhance signal synchronization across broader road networks.

In addition, the current prototype does not model complex real-world scenarios such as multiple simultaneous emergency vehicles, conflicting priority paths across multiple intersections, or rapidly changing traffic densities. Handling such cases requires more sophisticated coordination mechanisms, including multi-vehicle conflict resolution, dynamic flow-adaptive signal optimization, and large-scale traffic simulation. These aspects were intentionally beyond the scope of this proof-of-concept study but will be incorporated in future work to enhance robustness under congested or rapidly evolving traffic conditions.

Moreover, the current dataset augmentation relied on basic geometric transformations such as flipping, rotation, and scaling. Future work will incorporate more realistic augmentation methods—such as random brightness and contrast adjustment, Gaussian blur, and synthetic weather effects—to simulate lighting changes and reduced visibility in adverse conditions. These enhancements are expected to further improve detection robustness in real-world deployments.

Furthermore, the current evaluation focuses primarily on the prioritization of emergency vehicles and does not quantitatively assess the impact on regular traffic flow. The proposed system mitigates potential interference by restoring the original signal sequence after a short buffer period once the emergency vehicle passes. Future work will include detailed quantitative analysis of regular traffic metrics such as waiting time, queue length, and throughput to ensure balanced efficiency across all vehicle types.

While the proposed framework demonstrates clear reductions in emergency-vehicle delays, the present study does not quantify the impact of preemption on ordinary traffic flow. Giving priority to emergency vehicles can temporarily increase waiting time, queue length, and phase disruption for non-emergency vehicles, particularly in congested or high-demand intersections. Modeling these secondary effects requires multi-objective performance analysis and large-scale simulation of heterogeneous traffic streams, which were beyond the scope of this proof-of-concept evaluation. Future work will incorporate quantitative metrics for normal traffic, including average delay, queue formation, throughput, and signal recovery time, to more comprehensively assess system-wide performance.

Despite these limitations, the system serves as a practical proof-of-concept for real-time traffic control tailored to emergency scenarios. It lays a strong foundation for future extensions involving real-time video processing, predictive traffic modeling, and integration with existing smart transportation infrastructure.

VI. CONCLUSION

This paper presented an intelligent traffic light control system aimed at enhancing emergency response times by integrating deep learning-based emergency vehicle detection with real-time traffic signal optimization. The system employs a convolutional neural network for accurate vehicle classification and a dynamic signal adjustment strategy that prioritizes emergency vehicles across multiple intersections. Implemented as a web-based application, the framework offers a low-cost, scalable solution suitable for deployment in smart city environments.

Experimental evaluation showed that the proposed convolutional neural network model achieved a high classification accuracy of 97.0%, with reliable performance validated through standard metrics (precision, recall, F1-score, and accuracy) and confusion matrix analysis. The system demonstrated its ability to reduce delays for emergency vehicles by adjusting signal timing dynamically based on detection results and contextual factors such as speed and distance.

In future work, we plan to enhance the system by integrating GPS data and V2I communication, enabling better coordination in environments with occlusion or limited visibility. Additionally, incorporating real-time video stream processing and predictive tracking mechanisms will allow for proactive traffic signal preemption. Testing the system in live or simulated urban environments with varying traffic loads will further validate its robustness. Finally, adopting adaptive optimization or reinforcement learning algorithms

could improve the signal timing logic, allowing the system to learn and optimize over time based on traffic patterns.

Overall, the proposed solution demonstrates practical feasibility and serves as a foundation for next-generation traffic management systems that support intelligent and responsive emergency vehicle prioritization.

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